

Theory –last hour revision

Q1:- Give advantages & disadvantages of using computer rather than (nurses, students, etc).

Q2:- Give advantages & disadvantages of having laptop computer rather than a desktop computer.

Q3:- Give advantages & disadvantages of using robot rather than humans.

Q4:- Give advantages & disadvantages of using DTP rather than website.

Q5:- Give advantages & disadvantages of using GUI.

Q6:- Give advantages & disadvantages of using CLI.

Q7:- Give the differences between RAM and ROM.

Q8:- Give advantages & disadvantages of using fingerprint.

Q9:- Give the usage, advantages & disadvantages of the following:-

- a) Magnetic Stripe Reader.
- b) Chip & Pin readers.
- c) Light pen.
- d) OMR.
- e) OCR.
- f) MICR.

Q10:- Describe how a microprocessor-controlled such a system.

Q11:- Name three sensors and its uses in the following:-

- i. Washing machine.
- ii. Greenhouse.
- iii. Growing condition.
- iv. Weather station.
- v. Such a system.

Q12:- What is meant by EFTPOS and how such a system work?

Q13:- Why computer can't read data directly from sensor, and what device is needed to allow computer to read these data?

Q14:- Describe the facts of the microprocessor-controlled on lifestyle?

Q15:- Describe the inputs, outputs, hardware and processing involved in such a burglar alarm system?

Q16:- What is meant by video-conferencing, and give advantages & disadvantages of using it, and hardware needed for using it?

Q17:- Differences between hard disc and the following:-

- a) Floppy disc
- b) CD ROM

Q18:- Explain why the PCs is used with the following:-

- a) Pen drives
- b) Blu-ray
- c) Fixed hard disc drive
- d) Magnetic tape
- e) CD R
- f) DVD ROM

Q19:- Give the definition, and differences of the following:-

- a) WAN
- b) WLAN
- c) LAN

Q20:- Give the advantages & disadvantages of WLAN.

Q21:- Give the advantages & disadvantages of using internet.

Q22:- Give the differences between Intranet & Internet.

Q23:- Explain why connecting computers to network may need:-

- A Modem
- A browser.
- An ISP.
- Bridge.
- Switch.
- Hub.

Q24:- What should managers do to make their data secure?

Q25:- Describe the role of a proxy server?

Q26:- Describe the role of router?

Q27:- What are the features of the following:-

- i. Database.
- ii. Spreadsheet.

Q28:- Give the advantages of creating database rather than manual system.

Q29:- Give the Navigation aids that would be part of well-designed on screen-input form.

Q30:- What is meant by simulation?

Q31:- Give the advantages of sending message by e-mail rather than by post/hand.

Q32:- Give the advantages & disadvantages of using online shopping.

Q33:- Give the advantages & disadvantages of using mobile phone in accessing internet rather than PC

Q34:- What is meant by virus, and how they can be prevented?

Q35:- What are the purposes of Decryption & Encryption?

Q36:- What is meant software copyright, and how to prevent it from being broken, and how to break it?

Q37:- What is meant by Blog & Wiki?

Q38:- Give the types of security needed to prevent hacking.

Q39:- State health & safety problems, and the way to prevent them.

Q40:- State the ways in which computers containing personal data can be misused, and how to prevent them.

Q41:- Give the steps order of Chip and PIN system.

Q42:- Give reasons why some situations should be modeled.

Q43:- Give examples of computer situations.

Q44:- What is meant by expert system and what they might be used?

Q45:- Describe how an expert system can be created & evaluated?

Q46:- Describe the batch processing of bank cheques?

Q47:- Describe how an expert system diagnoses the faults?

Q48:- Give the components of a typical expert system?

Q49:- Give the methods of collecting information about a system.

Q50:- Explain the implementation types, and give advantages & disadvantages of each one of them?

Q51:- State what is meant by data validation & verification, and what is difference between them?

Q52:- State the types of evaluation, and its strategies with its reason.

Q53:- State the items of each document.

Q54:- What is meant by:-

- a) Normal range
- b) Abnormal range

c) Extreme range

Q55:- What is meant by backup, and why it is necessary, and why backing storage is needed, while computer systems have RAM?

Ans1:- Advantages:-

- Computer doesn't take breaks.
- Computers can work in any condition (even if dangerous).
- Computers can take more than one result in a moment.

- Computer takes data more accurately.
- Computers don't need to be paid.

Disadvantages:-

- Setting up is very expensive.
- Data can be lost (if not saved well).
- Maintenance cost can be expensive.
- Hackers may display/amend/delete the data.
- Unemployment.

Ans2:- Advantages:-

- Laptop can be easily transported.
- Laptop needs less space.
- Laptop can access it anywhere.

Disadvantages:-

- Laptop can be easily stolen.
- Laptop can be easily damage.
- Laptop needs charge from time to time.
- Laptop screen is not clear as the screen of desktop.

Ans3:- Advantages:-

- Robots can work 24 hours a day every day.
- Robots don't need to be paid a wage.
- Robots can work in any condition (even if it dangerous).

- Robots perform tasks more quickly.

Disadvantages:-

- Unemployment.
- Setting up is very expensive.
- Workers become deskilled.
- Maintenance cost can be expensive.

Ans4:- Advantages:-

- Any member of local family can see it without worrying.
- No need of internet to access it.
- Can read it anywhere.

Disadvantages:-

- No sound in it.
- No animations in it.
- Not as easy to update.

Ans5:- Advantages:-

- Can use mouse (no need of typing in commands).
- Easier to change/edit.
- Easier to open programs.

Disadvantages:-

- Bloated software can waste memory.
- Scripting a GUI interaction is not easily.
- Requires a higher threshold for the interface.

Ans6:- Advantages:-

- Requires a lower threshold for the interface.
- Scripting it is easily.
- No bloated software needed.

Disadvantages:-

- Typing in commands is difficult.
- Need to learn a lot of commands.
- No media.

Ans7:- RAM is random access memory, ***ROM*** is read only memory.

- **RAM** is temporary, **ROM** is permanent.
- **RAM** is volatile, **ROM** is non-volatile.
- **RAM** lost when power is lost, **ROM** not lost even if power is lost.

Ans8:- Advantages:-

- Easy to use.
- More accurate.
- Students will not be able to clock each other.

Disadvantages:-

- Expensive to install.
- Takes time to enter.
- If the finger is scratched, computer will not allow you to enter.

Ans9:-

Magnetic stripe reader:- Holds personal details.

Advantages:- - Faster entry.

- Easy to reprogram.
- Cards are inexpensive.
- Easy to use.

Disadvantages:- - Very limited storage capacity.

- Affected by magnet.
- Not very secure.
- Disable people find it hard

Chip & Pin readers :- Payment, identification cards.

Advantages:- Extra layer of security.

- Allow immediate updating.
- Not affected by magnet.
- Disable people find it easy.

Disadvantages:- Expensive to produce.

- Card holder needs to very careful to protect their PIN.
- If the customer forgets the PIN, transactions will be cancelled.

Light pen:- Allow the user to point to displayed objects, or to draw on screen.

Advantages:- - Make you able to interact with screen.

- Better suited to artistic.
- Light pen systems is very efficient for situations.

Disadvantages:- Light pen system need very specified screen.

- Light pens require somewhat awkward positioning.
- Inefficient for general use.

OMR:- To input data from pencil/ pen.

Advantages:- Quick entry method.

- Easy for user to fill in the form.
- More accurate.

Disadvantages:- If marks didn't fill completely, it may not be read correctly.

- Only suitable for recording one out of selection of answers.

OCR:- Convert scanned image to understandable text.

Advantages:- Faster method of entry.

- Disable people find it easy.

Disadvantages:- A lot of errors are produced.

- Cannot read hand writing
- Requires very high quality scanned image.

MICR:- Uses magnetic ink to print routing and account numbers.

Advantages:- Ease of readability.

- High security.
- Fast way to read bank data.

Disadvantages:- Expensive.

- There is a limit to the character set that can be used.

Ans10:- Microprocessor stores required temperature/water level as preset value.

- Data is converted from analogue to digital/ ADC is used.
- Compares temperature/water level with pre-set value.
- If temperature is lower than preset value microprocessor sends a signal to turn heater on.
- If higher than preset value microprocessor sends a signal to turn heater off.
- If values are equal microprocessor does nothing.

Ans11:-

Sensors	Washing machine.	Greenhouse	Growing condition	Weather station.
<i>Temperature</i>	√	√	√	√
<i>pH</i>	√	√	√	X
<i>Pressure</i>	√	X	X	√
<i>Light</i>	X	√	√	√

Ans12:- - - Electronic Funds Transfer at Point of Sale.

- Enables payment for goods at a checkout using credit.
- Goods are purchased and bill is calculated.
- Customer inserts card into chip reader.
- PIN is entered.

- PIN is compared with that stored on the chip.

Ans13:- Sensors produce analogue signals, while computer read only digital signals, so analogue to digital converter is needed.

Ans14:- - People become lazier.

- Can go out more. / - More free time.

Ans15:- Inputs:- Sound, Temperature and light sensors.

Outputs:- Alarm light flashes. / Alarm sounds

Processing:- Microprocessor continually monitors sensors.

If light sensor detects light beam interrupted.

If temperature sensed by processor is greater than preset value.

If sound sensed by processor is greater than preset value.

Microprocessor sends signal to output device.

Ans16:- Video conference is a live connection between people in separate locations, in purpose of communications.

Hardware:- Webcam – Microphone – Speakers.

Advantages:- - No need of paying for conference room.

- No need of paying for employee travelling expenses.
- Conference can be called at short notice.

Disadvantages:- - Lack of personal touch.

- Visual and sound may be not clear.
- Difficult to have international meeting due to time differences.

Ans17:-

(a) - Hard discs are non-portable, while floppy discs are portable.

- Hard discs hold more data than floppy discs.
- Hard discs have faster data access times than floppy discs.
- Hard discs have faster data transfer rates than floppy discs.

(b) - Cannot be moved easily from computer to another.

- Hard discs have faster data access times than CD ROM.
- Hard discs have faster data transfer rates than CD ROM.
- Hard discs hold more data than CD ROM.

Ans18:-

Pen Drives:- - More robust than memories / - Carrying software files.

Blu-ray:- - Used to store films. / - Large capacity

Fixed hard disk:- - Stores OS / - Stores software.

Magnetic tape:- - Make backups. / - Batch Processing.

CD RW:- - Rewritable disc. / - Recording television programs.

DVD/ CD ROM:- Display movies

Ans19:-

WAN:- Is a computer network that covers a broad area.

LAN:- Is a computer network that covers a small area.

<u>Differences</u>		
<u>WAN</u>	<u>LAN</u>	<u>WLAN</u>
Wide Area Network.	Local Area Network.	Wireless Local Area Network.
Low transfer rate.	High transfer rate.	Use wireless to transfer data.
Can't connect WAN to WAN.	Can connect LAN to LAN.	
	Greater security.	Less security.

Ans20:- Advantages:-

- Easier to add/move workstations.
- Easier to provide connectivity in difficult areas.
- Can access network from anywhere.
- Installation is fast and easy.

Disadvantages:-

- Long term cost benefits are harder to achieve.
- Lower wireless bandwidth.
- Security is more difficult to guarantee.
- Devices will be operated at a limited distance.

Ans21:- Advantages:-

- Can share information.
- Can share hardware & software.
- Can play games.
- Can communicate with others (by email).

Disadvantages:-

- Work can be misused.
- Greater risk of hackers.
- Viruses can be easily spread.
- Cost of extra equipments.

Ans22:-

<u>Internet</u>	<u>Intranet</u>
International Network.	Internal Restricted Access Network.
Is WAN.	Is LAN.
Can access it anywhere.	Password controlled pages.
Is global.	Is within one organisation.

Ans23:-

A Modem:- To convert analogue signals to digital signals.

A browser:- To access the (world wide) web.

An ISP:- To access the internet.

A Bridge:- Connects networks.

A Switch:- Directs data packets to specific computers.

A Hub:- Broadcasts data packets to computers in a LAN.

Ans24:-

Firewall:- Prevents unauthorised computers having access to the network.

Antivirus:- Removes viruses from storage media.

Encryption:- Data becomes unreadable to unauthorised users.

Ans25:-

- Act as a web server.
- Act as buffer.
- Can store the web pages.
- Can block certain sites.

Ans26:-

- Connect LAN to WAN.
- Connect LAN to internet.
- Reduce collisions.
- Passes data to switch.

Ans27:- Database:-

- Easier to edit data.
- Less storage capacity is used.
- Quicker to enter new data.

Spreadsheet:-

- Performs what ifs.
- Replicate formulas.
- Can make graphs or charts.

Ans28:- - Quicker to enter/update/edit data.

- Few errors are likely
- Less duplication of data

Ans29:- - Backward

- Forward
- New record button
- Exit button

Ans30:- Simulation modeling is the process of creating and analyzing a digital prototype of a physical model to predict its performance in the real world

Ans31:- - Message can be send anytime.

- Message will be delivered faster.
- You don't have to leave your house to send email.
- Message can be read from any location.
- You can send attachments.

Ans32:- Advantages:-

- Don't have to leave the house.
- Saves travelling time.
- Saves travelling expenses.
- Can shop when shops are closed.

Disadvantages:-

- Need a credit card.
- Need internet and computer.
- Less of personal touch.
- Can't see the goods in reality.

Ans33:- Advantages:-

- Mobile is more portable than laptop.
- Can access internet from greater number of places.
- Easier to use while on the move.

Disadvantages:-

- Display screen is smaller.
- Keyboard is smaller.
- Some sites do not have full facilities for mobile phone.

Ans34:-

- A program that has been created with the purpose of damaging a computer system.
- Prevent entering suspicious websites.
- Install good anti-virus software.
- Stop downloading software.
- Install anti-malware software.

Ans35:-

Encryption:- Prevents hackers from understanding data.

- Scrambling the data.
- It doesn't destroy viruses.

Decryption:- The purposes of decoding data.

- Requires a secret key or password.

Ans36:-***Prevention:-***

- Should not be copied.
- Should not be used on network.
- Should not be renting without owner's permission.

Broken by:-

- Giving software copies to others without the owner's permission.
- Making a copy of a disk without the owner's permission.
- Selling copies of the software without the owner's permission.

Ans37:-***Blog:-*** Personal Forum.

- Usually one author.
- Can be an electronic diary.
- Others can comment.
- Only author can edit entries.

Wiki:- General forum.

- Usually Many authors.
- Usually objective.
- Internal and external links.
- Contributors can edit entries.

Ans38:-

Proxy server:- Block suspicious websites.

Firewall:- Provide a shield between a computer and the internet.

Data Encryption:- Prevent hackers from understanding data.

Parental guidance settings:- Can block undesirable topics.

Ans39:-

<i>Safety</i>	<i>Precaution</i>	<i>Healthy</i>	<i>Precaution</i>
Electrocution.	Cable insulation.	Headaches.	Take regular breaks.
Trailing cables.	Create ducts.	RSI.	Finger exercises.
Fire.	Extinguisher.	Eyestrain.	Get anti-glare screen.
Equipment falling.	Large tables.	Backaches.	Straight back chair.

Ans40:-

- Hackers may read the data and pass it on.
- Hackers may delete the data.
- Hackers may amend the data.

Prevention:-

- Data should be encrypted.
- Use username and password.
- Do not connect the computers to a network.
- Physical security (locks on computer room doors).

Ans41:-

- System used to read bank/credit cards.
- At ATM or EFTPOS systems.
- PIN is typed in using PIN PAD.
- Details on chip are read by chip reader.
- If PIN correct the customer will be able to enter the account.
- If not correct, the system will ask the customer to re-enter the PIN.

Ans42:-

- Some situations is too dangerous to build.
- Some situations is too expensive to build.
- Some situations is time consuming to build.
- Some situations its time scales are too vast.

Ans43:-

3D modelling. / Flight simulation. / Driving simulation.

Ans44:-

- Expert systems are special databases which are designed to copy the expertise and knowledge of a human expert in many subjects.
- **Used in:-** Car fault diagnosis. / Tax. / Chess games.

Ans45:-

- Interview the experts, and collect knowledge.
- Knowledge base is created, organized and stored.
- Inference engine is created, organized and stored.
- Rule base is created, to allow users to communicate with the expert system.
- Testing by questions in form of (check boxes).

Ans46:-

- Cheques are collected together, during the course of the day.
- Cheques are then processed all at once.
- Cheques are processed overnight.
- Bank accounts updated following morning.

Ans47:-

- Data is read by sensors.
- Uses interactive interface.
- Expert system compares symptoms, with those in the knowledge base, using rules base, then matches are found.

Ans48:-

- Symptoms are entered using the user interface
- User interface displays questions.
- Expert system compares symptoms, with those in the knowledge base, using rules base, and then matches are found.

Ans49:- Interviews – Questionnaires – Observations – Examining documents.

Ans50:-

Direct implementation:- Means that the old system is stopped, and the new system is started.

Advantages:- Save time. / Save costs.

Disadvantages:- Staff may find it difficult

- If the system is down there is no other system to act as backup

Phased implementation:- Means that the new system is introduced step by step (in phases)

Advantages:- If the system is down there other will act as a backup

- Easy to train staff.

Disadvantages:- A problem is that it might take too much time to introduce the new system.

- Expenses of running 2 systems.

Pilot implementation:- Means that the new system replaces the old system in one operation, but only on a small scale.

Advantages:- Saves cost. / Easy to train staff.

Disadvantages:- It will take too much time to get a pilot to completely replace of the old system.

Parallel implementation:- Means that the new system is started, but the old system is kept in running (in parallel)

Advantages:- Easy to train staff.

- If the system is down there other will act as a backup.

Disadvantages:- Expenses of running 2 systems.

- A problem is that all tasks should be done twice.

Ans51:-

Data Validation:- It checks if the data from the correct type or not.

Range, presence, digit, data type, length, and format checks.

Data verification:- It makes sure that data is accurate.

- Proof reading, double-entry.

<u>Differences</u>	
<u>Data Validation</u>	<u>Data Verification</u>
Checking if the data is reasonable or acceptable.	Checking if data has been accurately copied or not.
It will pick up errors if verification does not	It will pick up errors if validation does not
It always carried out by computer	It always carried out by user

Ans52:-

- Ease of use
- The appropriateness of the solution.

<u>Strategy</u>	<u>Reason</u>
Comparing the solution with the original task requirements.	To ensure that requirements have been met.
Collect data from the users of the new system.	To see how well the new system works.
Analyse the user's responses to the testing results	So that users needs are takes into account

Ans53:- *User Documentation*

- How to load/run software.
- How to save a file.
- How to Print.
- How to add/edit/delete records.

Technical documentation

- Program listing.
- File structure.
- Print formats
- Purpose of the system.
- Screen layouts.

Ans54:-

Normal range:- Data within the given range.

Abnormal range:- Data outside the given range.

Extreme range:- Data on the boundaries of the range.

Ans55:- Backup is the activity of copying files or databases, in case the file is accidentally deleted, corrupted, changed, deleted, hacked, or got a virus. It is necessary even if RAM is present because, if power is lost RAM will be lost, while backing up the files will not be lost even if power is lost.